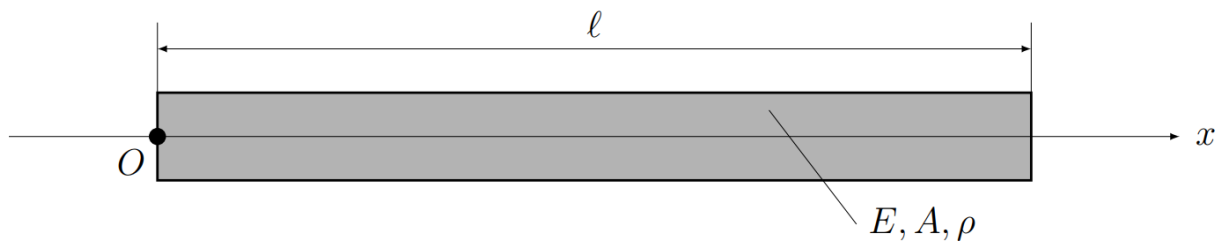




ME-473
MINI-PROJECT 1

Natural frequencies and mode shapes of a uniform bar



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1 Introduction

In this report, we are interested in studying the natural frequencies and mode shapes of a 1D longitudinal metal bar clamped at one end using FEA. More specifically, quadratic linear elements are used along with the local form of the Galerkin method to determine the natural frequencies of the bar.

2 Methodology

2.1 Material and geometrical properties

The geometric and material quantities used in the analysis are as follows.

- $\ell = 1$ [m] : total length of the bar
- $A = 0.01$ [m²] : uniform cross-section of the bar (the shape is irrelevant)
- $E = 210$ [GPa] : Young's modulus of steel
- $\rho = 7850$ [kg/m³] : mass density of steel

2.2 Governing equation and boundary conditions

The strong form of the equation governing the deformation of the 1D bar is given below, where the boundary conditions are set with zero displacement at the clamped end of the bar ($x = 0$) and zero stress at the free end of the bar ($x = \ell$).

$$\begin{cases} EA\partial_{xx}^2 u(x, t) = \rho A\ddot{u}(x, t), & x \in [0, L] \\ u(0, t) = 0 \\ \partial_x u(\ell, t) = 0 \end{cases} \quad (1)$$

The weak form of the equation follows:

$$\int_0^\ell EA\partial_x u_1^h \partial_x (\delta u_1^h) dx + \int_0^\ell \rho A\ddot{u}_1^h \delta u_1^h dx = 0 \quad (2)$$

With initial conditions and solution space as described below:

$$\begin{aligned} \int_0^\ell \rho A u(x, 0) \delta u_1(x) dx &= \int_0^\ell \rho A u_0(x) \delta u_1(x) dx \\ \int_0^\ell \rho A \dot{u}(x, 0) \delta u_1(x) dx &= \int_0^\ell \rho A v_0(x) \delta u_1(x) dx \\ \mathcal{U} &= \{u_1(\cdot, t) \in H^1(]0, \ell[) \mid u_1(0, t) = 0 \forall t \in]0, T[\} \\ \mathcal{V} &= \{\delta u_1 \in H^1(]0, \ell[) \mid \delta u_1(0) = 0\} \end{aligned} \quad (3)$$

2.3 Finite element method

We first start by discretizing the bar into n elements where $n = \{1, 2, 4, 8, 16\}$. n is varied to study the impact of the number of elements on the convergence of the natural frequencies. The total number of nodes is simply:

$$n_{nodes} = 2n + 1$$

as the bar is made up of quadratic elements which contain 3 nodes each.

Since the local method is used for the finite element analysis, a connectivity table must be created to relate the numbering of the archetypal element to the global system. The local nodes are numbered in order 1, 3, 2 (from left to right) by convention for quadratic elements, and the global nodes are numbered 1 to $2n + 1$ from left to right. We can create a $n \times 3$ matrix which contains the global number of each node for each element. An example with 4 elements is shown in Table (1).

Element	1	2	3
$^1\Omega$	1	3	2
$^2\Omega$	3	5	4
$^3\Omega$	5	7	6
$^4\Omega$	7	9	8

Table 1: Connectivity table T for quadratic elements mapping each local node to their respective global node with $n = 4$.

In general, the table or matrix can be built in the following way:

$$T_i = (2i - 1, 2i + 1, 2i) \quad \forall i = 1, \dots, n$$

where T_i is the i -th row of the connectivity table T .

The Galerkin method transforms Equation (2) into a discrete set of equations through the use of a displacement approximation, defined as follows:

$$\begin{aligned}
 u(x, t) &\approx u^h(x, t) \in \mathcal{U}^h \subset \mathcal{U} \\
 u^h(x, t) &= H(x) q(t) \\
 \delta u(x, t) &\approx \delta u^h(x, t) \in \mathcal{V}^h \subset \mathcal{U} \\
 \delta u^h(x, t) &= H(x) q(t)
 \end{aligned} \tag{4}$$

In the local method, shape functions $H(x)$ are defined for a single archetypal element (${}^e\Omega$) and used for all elements on the domain using a coordinate change to determine the mass and stiffness matrices. The quadratic base functions eh_1 , eh_2 and eh_3 obey the Kronecker property ${}^ah_i(\xi_j) = \delta_{ij}$ with ξ as the local abscissa of the archetypal element. These are created using Lagrangian polynomials, and are plotted in Figure (1).

$${}^ah_1(\xi) = \xi(\xi - 1)/2, \quad {}^ah_2(\xi) = \xi(\xi + 1)/2, \quad {}^ah_3(\xi) = 1 - \xi^2 \quad \forall \xi \in [-1, 1]$$

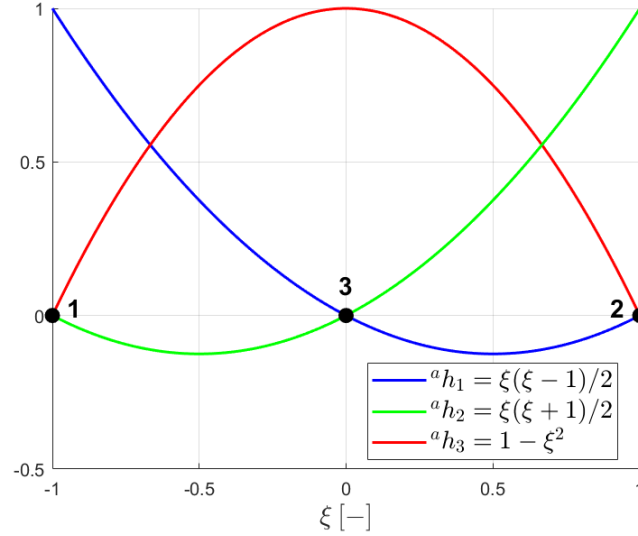


Figure 1: Quadratic base functions of the archetypal element with the 1,3,2 naming convention.

The derivatives of the shape functions used in the computation of the stiffness matrix K are simply:

$$\frac{d({}^ah_1)}{d\xi} = \xi - \frac{1}{2}, \quad \frac{d({}^ah_2)}{d\xi} = \xi + \frac{1}{2}, \quad \frac{d({}^ah_3)}{d\xi} = -2\xi \quad \forall \xi \in [-1, 1]$$

Equation (2) can be written using the displacement approximation as:

$$\mathbf{M}\ddot{\mathbf{q}} + \mathbf{K}\mathbf{q} = 0 \quad (5)$$

Where the global mass and stiffness matrices ($\mathbf{M}$ and $\mathbf{K}$) are created from the superimposed combination of elemental mass and stiffness matrices eM and eK . These matrices are defined for each global element as follows:

$${}^eM = \int_{x_1}^{x_2} \rho A H^T H dx, \quad {}^eK = \int_{x_1}^{x_2} EA \left(\frac{dH}{dx} \right)^T \left(\frac{dH}{dx} \right) dx \quad (6)$$

where x_1 and x_2 are the coordinates of the leftmost and rightmost nodes of the global element respectively.

A linear mapping between the archetypal element and the global element is defined to transform the local ξ coordinate to the global x coordinate:

$$x = \frac{1-\xi}{2}x_1 + \frac{1+\xi}{2}x_2 \quad (7)$$

Taking the derivative with respect to ξ :

$$\frac{dx}{d\xi} = J = \frac{x_2 - x_1}{2} = \frac{l_e}{2}$$

We obtain the relation $dx = Jd\xi$ where J is the Jacobian of the coordinate transformation and l_e is the length of the local element. We can re-write the stiffness matrix eK in its archteypal element form as:

$$\begin{aligned} {}^eK &= \int_{x_1}^{x_2} EA \left(\frac{dH}{dx} \right)^T \left(\frac{dH}{dx} \right) dx = \int_{-1}^1 EA \left(\frac{dH}{d\xi} \frac{d\xi}{dx} \right)^T \left(\frac{dH}{d\xi} \frac{d\xi}{dx} \right) J d\xi \\ &= \int_{-1}^1 EA \left(\frac{dH}{d\xi} J^{-1} \right)^T \left(\frac{dH}{d\xi} J^{-1} \right) J d\xi = EA \frac{2}{l_e} \int_{-1}^1 B^T B d\xi \end{aligned}$$

In the same way, the local mass matrix eM can be rewritten more simply as:

$${}^eM = \int_{x_1}^{x_2} \rho A H^T H dx = \int_{-1}^1 \rho A H^T H J d\xi = \rho A \frac{l_e}{2} \int_{-1}^1 H^T H d\xi$$

We are now able to assemble the global mass and stiffness matrices M and K using the following definition:

$$M = \bigcup_{e=1}^n {}^eM, \quad K = \bigcup_{e=1}^n {}^eK \quad (8)$$

Let's take $n = 2$ as an example. First, the local mass and stiffness matrices eM and eK are computed (eM and eK are independent of n):

$${}^eM = \frac{\rho A l_e}{30} \begin{pmatrix} 4 & -1 & 2 \\ -1 & 4 & 2 \\ 2 & 2 & 16 \end{pmatrix}, \quad {}^eK = \frac{EA}{3l_e} \begin{pmatrix} 7 & 1 & -8 \\ 1 & 7 & -8 \\ -8 & -8 & 16 \end{pmatrix}$$

The global mass and stiffness matrices M and K are now assembled by adding each contributing term at the correct location by looking at the connectivity table:

$$M = \frac{\rho A l_e}{30} \begin{pmatrix} 4 & 2 & -1 & 0 & 0 \\ 2 & 16 & 2 & 0 & 0 \\ -1 & 2 & 4+4 & 2 & -1 \\ 0 & 0 & 2 & 16 & 2 \\ 0 & 0 & -1 & 2 & 4 \end{pmatrix}, \quad K = \frac{EA}{3l_e} \begin{pmatrix} 7 & -8 & 1 & 0 & 0 \\ -8 & 16 & -8 & 0 & 0 \\ 1 & -8 & 7+7 & -8 & 1 \\ 0 & 0 & -8 & 16 & -8 \\ 0 & 0 & 1 & -8 & 7 \end{pmatrix}$$

The boundary conditions can now be applied. This results in the elimination of the first row and first column of each matrix as the virtual displacements δu and real displacements u are fixed and equal to 0 at the first node. In the case $n = 2$, the reduced global mass and stiffness matrices become:

$$\mathbf{M} = \frac{\rho A l_e}{30} \begin{pmatrix} 16 & 2 & 0 & 0 \\ 2 & 8 & 2 & -1 \\ 0 & 2 & 16 & 2 \\ 0 & -1 & 2 & 4 \end{pmatrix}, \quad \mathbf{K} = \frac{EA}{3l_e} \begin{pmatrix} 16 & -8 & 0 & 0 \\ -8 & 14 & -8 & 1 \\ 0 & -8 & 16 & -8 \\ 0 & 1 & -8 & 7 \end{pmatrix}$$

We are now able to compute the first $2n$ natural frequencies and mode shapes by solving the eigenvalues problem of the following equation:

$$\det(\mathbf{K} - 4\pi^2 f^2 \mathbf{M}) = 0 \quad (9)$$

where $\mathbf{M}$ and $\mathbf{K}$ are the reduced mass and stiffness matrices respectively and f is the vector containing the first $2n$ natural frequencies.

3 Results and Discussion

Table (2) shows the first five natural frequencies found both analytically and using the finite element method, for various numbers of elements used. Note that not all modes can be found with all numbers of elements. For n quadratic elements there exist $2n + 1$ nodes which can be seen as degrees of freedom, of which one is clamped and is defined as having a displacement of zero (and which therefore eliminates a row and column in the matrices used to solve for natural frequencies). Consequently, for n elements, we can only find the first $2n$ natural frequencies and mode shapes, as the $\mathbf{M}$ and $\mathbf{K}$ matrices used to solve the eigenvalue problem are of size $2n \times 2n$.

The analytical solution used to evaluate the natural frequencies f of the k -th order are given in Equation (10). Although the general solution to the 1D wave equation includes both odd and even terms (both $\cos(x)$ and $\sin(x)$ terms are present), our solutions preserve only the odd terms to satisfy the boundary conditions $u(0, t) = 0$ and $\partial_x u(L, t) = 0$.

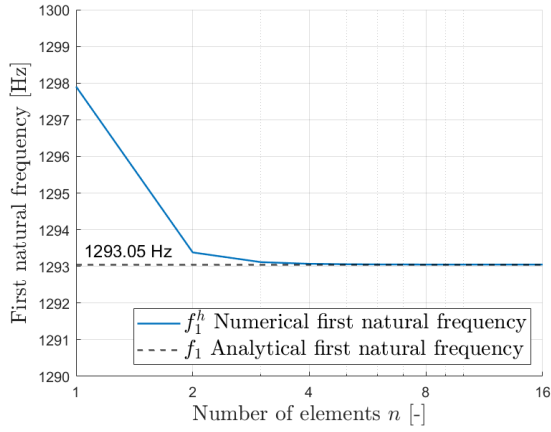
$$u(x, t) = \sum_{k=1}^{\infty} A_k \cos\left(\frac{(2k-1)\pi x}{2L}\right) \cos(2\pi f_k t + \phi_k), \quad f_k = \frac{2k-1}{4\ell} \sqrt{\frac{E}{\rho}} \quad \forall k \in \mathbb{N} \quad (10)$$

where the different A_k and ϕ_k terms depend on the initial conditions.

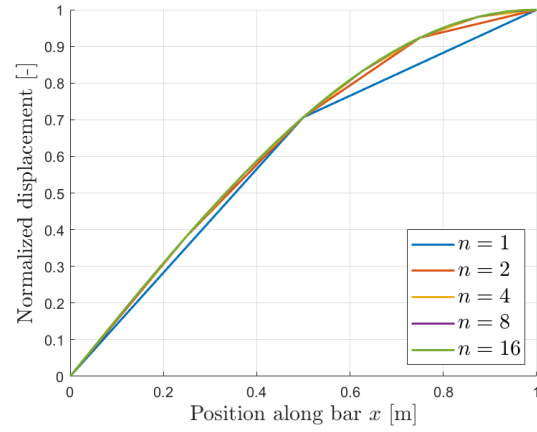
Figures (2 - 6) show the approximated natural frequencies and the 1D wave mode shapes as a function of the number of elements used. Notably, only the odd harmonics are shown, as discussed previously. As the amplitudes of the mode shapes depend on the initial conditions, their maximum values were arbitrarily normalized to 1.

Mode	Analytic	n = 1	n = 2	n = 4	n = 8	n = 16
1	1293.05	1297.90	1293.38	1293.07	1293.05	1293.05
2	3879.15	4670.74	3943.21	3884.99	3879.46	3879.17
3	6465.24	~	7226.34	6520.95	6469.22	6465.50
4	9051.34	~	11603.59	9293.83	9071.79	9053.71
5	11637.44	~	~	12654.70	11705.40	11642.17

Table 2: Exact and approximate natural frequencies (in Hz) found for various numbers of elements.

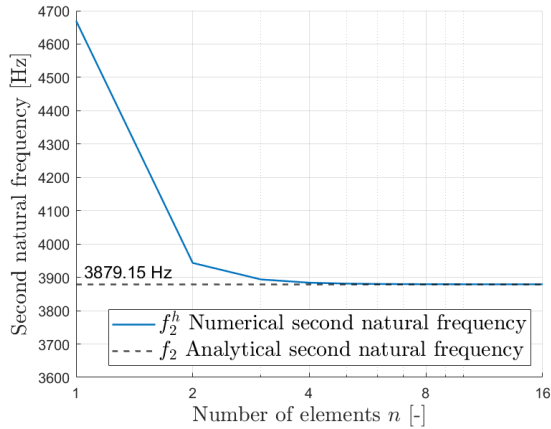


(a) Evolution of the first natural frequency.

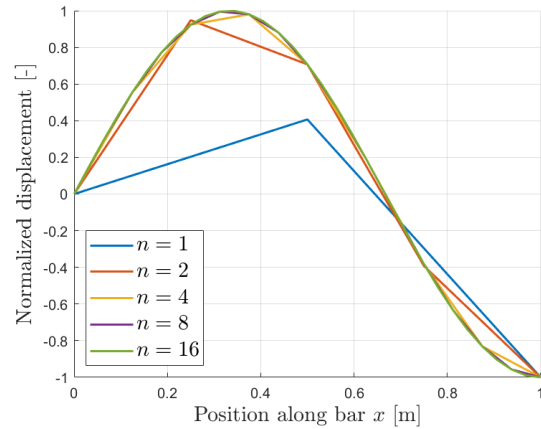


(b) Evolution of the first mode shape.

Figure 2: Illustration of the convergence of the first natural frequency f_1 with respect to the number of elements n .

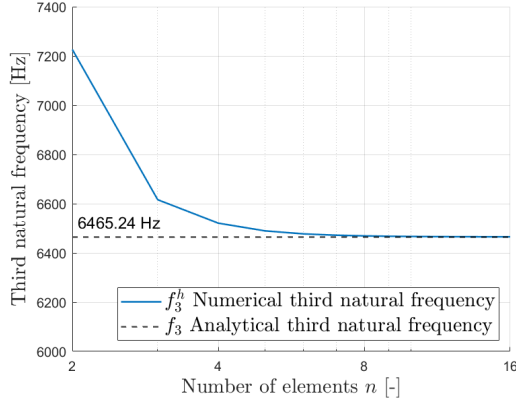


(a) Evolution of the second natural frequency.

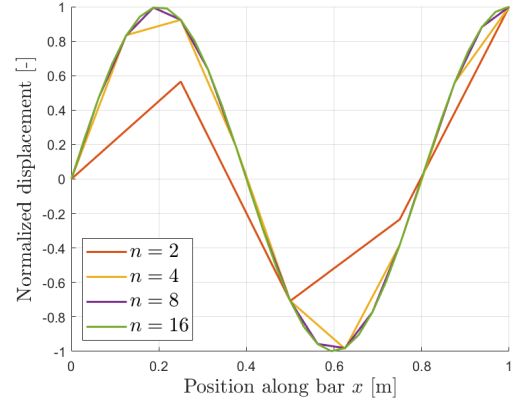


(b) Evolution of the second mode shape.

Figure 3: Illustration of the convergence of the second natural frequency f_2 with respect to the number of elements n .

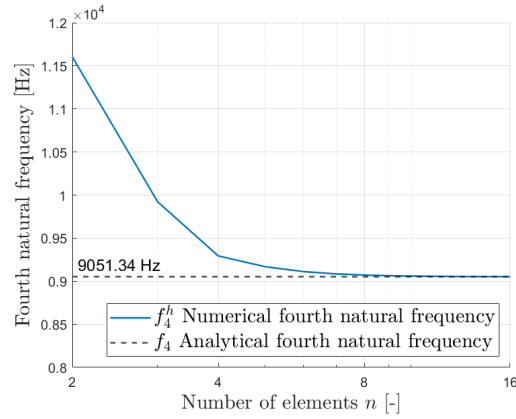


(a) Evolution of the third natural frequency.

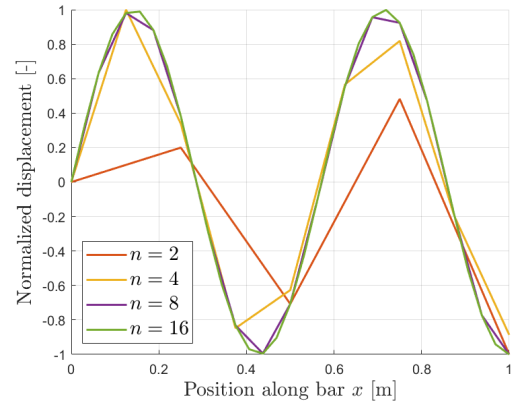


(b) Evolution of the third mode shape.

Figure 4: Illustration of the convergence of the third natural frequency f_3 with respect to the number of elements n .

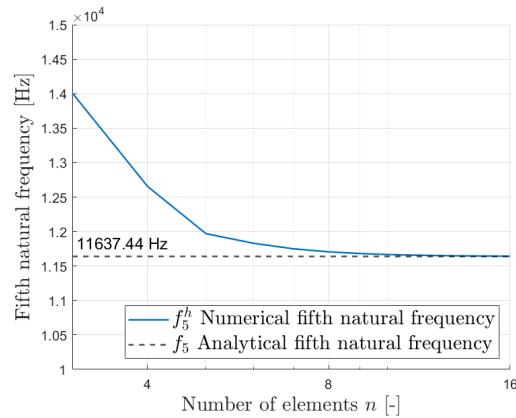


(a) Evolution of the fourth natural frequency.

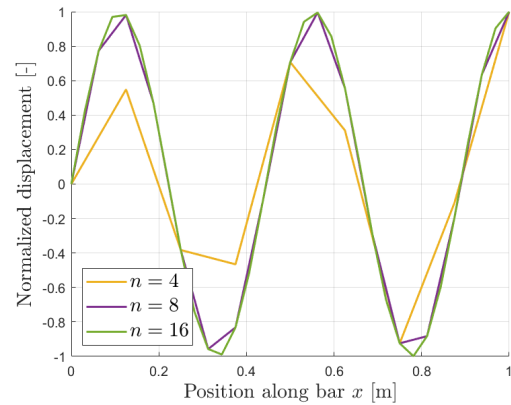


(b) Evolution of the fourth mode shape.

Figure 5: Illustration of the convergence of the fourth natural frequency f_4 with respect to the number of elements n .



(a) Evolution of the fifth natural frequency.



(b) Evolution of the fifth mode shape.

Figure 6: Illustration of the convergence of the fifth natural frequency f_5 with respect to the number of elements n .

In general, we observe that the natural frequencies converge rapidly as the number of elements n increases. However, the accuracy of the mode shapes, particularly for higher-order modes, is significantly affected when n is small. This occurs because the number of degrees of freedom (linked to the number of nodes) is insufficient to capture the increasingly complex oscillatory behavior of these higher-order mode shapes.

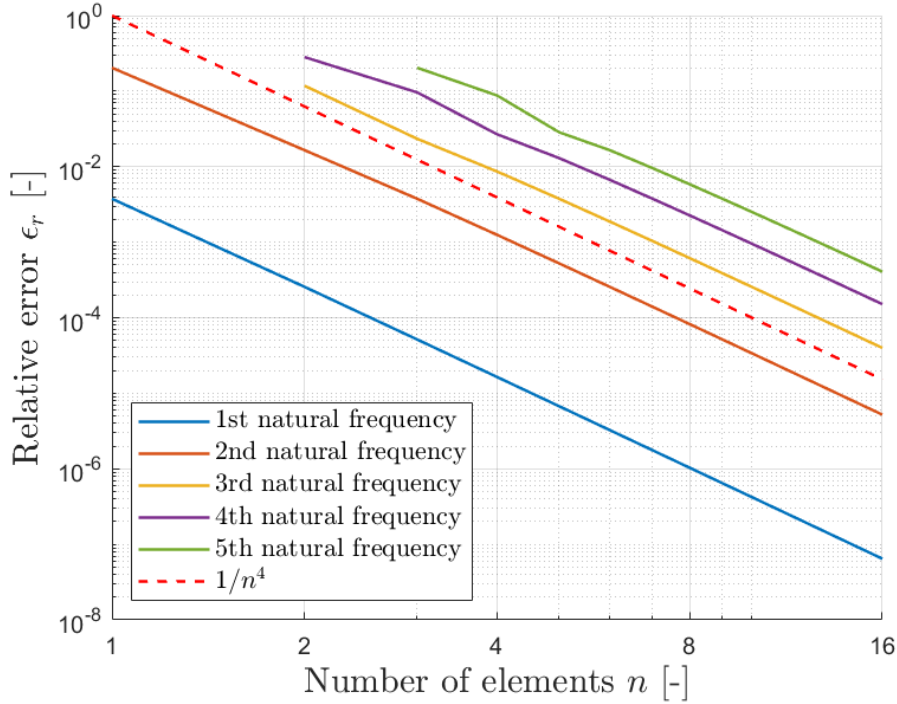


Figure 7: Evolution of natural frequency relative errors with respect to the number of elements n .

Figure (7) displays the change in relative natural frequency error with number of finite element used, for all investigated frequency modes. It is easy to see that the relative error is higher with higher natural frequency orders. More interestingly, we can observe how the relative error ϵ_r converges at the same rate as $1/n^{-4}$ for all natural frequencies. This means that the order of the error is $\mathcal{O}(h^4)$ where h is the characteristic size of the finite elements. To understand the origin of this error limit, we consider the solution of a 1D eigenvalue problem which uses polynomial shape functions. The error in the computed eigenvalues follows the relation:²

$$|\lambda^{(k)} - \lambda_h^{(k)}| = \mathcal{O}(h^{2p})$$

with p being the degree of the shape function used, $\lambda_h^{(k)}$, the approximated k -th eigenvalue, and the real value, $\lambda^{(k)}$. In our case $p = 2$, which yields an error convergence rate of $\mathcal{O}(h^4)$. Of note, the relative error in the eigenvalues and the natural frequencies exhibit the same convergence behavior, despite a difference of squares between the two.

4 Conclusion

This study has analyzed how the number of elements used in the local finite element method impacts the approximated natural frequencies of a clamped 1D bar, for the case of quadratic shape functions. Our results show that as the number of elements increases, the computed frequencies converge rapidly to the analytical solution. However, higher-order mode shapes require a larger number of elements for an accurate representation. Additionally, a number of n elements allows only the first $2n$ natural frequencies to be calculated, meaning not all discretizations are capable of producing all natural frequencies. The observed convergence of error of the natural frequencies follows $\mathcal{O}(h^4)$, consistent with theoretical predictions for polynomial shape functions of degree $p = 2$. This study thus demonstrates the importance of the choice of domain discretization in achieving accurate results in the analysis of vibrational finite elements.

5 References

1. *Stefano Burzio*, Dynamic finite element analysis of structures [ME-473]. *EPFL* (2025)
2. Remark 2.1, *Daniele Boffi*, Finite element approximation of eigenvalue problems, *Acta NumeTica* (2010), pp. 1-120, (doi: 10.1017/80962492910000012)